

Shubham Hadawle

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EDUCATION

Northeastern University (Khoury College of Computer Sciences), Boston

Expected Dec 2027

Master of Science in Computer Science

Relevant Courses: Programming Design Paradigms, Database Management Systems, Algorithms, ML Ops, NLP

University of Mumbai, Mumbai

July 2020 - June 2024

Bachelor of Engineering in 'Artificial Intelligence and Data Science'

Relevant Courses: Machine Learning, Artificial Intelligence, Python, Data Structures, Object Oriented Programming – Java, Deep Learning, Data Analytics & Visualization, Probability and Statistics

TECHNICAL SKILLS

- Languages: Python, Java, C, C++, JavaScript, R, Shell Scripting
- Databases & Big Data: SQL, MySQL, NoSQL, MongoDB, PostgreSQL, Firebase, Apache Spark, Hadoop
- ML Libraries: Numpy, Pandas, Matplotlib, Seaborn, Plotly, PyTorch, Scikit-Learn (Sklearn), TensorFlow, Keras, PySpark, Scipy, OpenCV, Flask, NLTK, SpaCy, LangChain
- Mobile / Web Dev: XML, Android Studio, HTML5, CSS, NodeJS, Express.js, React, Streamlit
- Tools & Technologies: Git, GitHub, AWS (EC2, S3), Google Cloud Platform, Microsoft Azure, Flask, Postman, Docker, REST APIs, Hugging Face, LLMs (Large Language Models), XGBoost, Linux, Unix, MATLAB
- Other: Deep Learning, Natural Language Programming, Computer Vision, Image Processing, Time Series Analysis, OpenCV, Generative-AI, Transformers, ETL Pipeline, EDA (Exploratory Data Analysis), Data Modeling, Data Processing, Linear Algebra

PROFESSIONAL EXPERIENCE

Motilal Oswal Financial Services | Software Engineer - Machine Learning Intern

Sept 2025 – Nov 2025

- Improved retrieval quality for internal **RAG (Retrieval-Augmented Generation)** system by developing a **clustering pipeline** for Transformer-based text **embeddings** (SBERT) using **hierarchical clustering (HAC)** algorithm.
- Leveraged strong **problem solving** and **debugging skills** to automate the processing of large-scale financial research PDFs, to generate accurate, context-preserving document clusters with 72% chunk coverage.
- Contributed to internal AI initiatives, enhancing the accuracy of RAG pipelines by engineered a highly scalable **document chunking** and **clustering system** integrating **OCR** parsing workflows & LandingAI's **Agentic Document Extraction (DPT-2)** for PDF parsing.

Axis Bank, Mumbai, India | AI / ML Engineer Intern

May 2025 – Aug 2025

- Devised a **GraphRAG** retrieval system integrating knowledge graphs with LLMs for **scalable** information retrieval.
- Built end-to-end **Text-to-SQL** infrastructure using **Neo4j**, to help map **natural language queries to SQL** on financial reports.
- Reduced manual querying overhead on large-scale customer datasets by leveraging structured schema graphs.

Bhabha Atomic Research Centre (BARC), Mumbai, India | Software Engineer Intern - Data Science

Nov 2023 – Jan 2024

- Designed Deep Neural Networks (**LSTMs, Autoencoders, VAEs and Neural ODEs**) to solve Point Kinetic Equations that simulate the behavior of the Uranium-235 nuclear reactor.
- Utilized **Python** frameworks like **Numpy, Pandas, Matplotlib, Seaborn, Pyplot, Scikit learn, TensorFlow, Keras** and **PyTorch** (torchdyn/torchdiffeq) to automate the process of mathematical calculations thereby increasing calculation efficiency.
- Optimized cross-domain adaptation using **Adam optimizer** to refine the **loss function**, reducing the loss from **1.19520 to 0.00062**.

Stocc Guru, Mumbai, India | Software Development Engineer Intern

July 2021 – Sept 2021

- Developed a full-stack **Java Android app** with client-server architecture using **Java, Python, Flask, and REST APIs**, serving real-time analytics to 1,000+ users.
- Integrated **CI/CD** pipelines via **GitHub Actions** for automated builds, testing, and deployment.
- Implemented backend system following **SOLID Principles**, reducing API response latency by 2.4% via optimized **JSON** processing.

PROJECTS AND PAPERS

Platformer Game: Where is my Duck?

([Link to Mobile Game](#) / [Link to Desktop App](#))

- Engineered and **published** a 2D platformer game for both **mobile** and **desktop (Windows/Linux/ macOS) apps** using GDevelop5, achieving **4.7/5 stars average rating** across both platforms with **1000+ downloads** on **Google Play Store**.
- Monetized the Android app** via **Google AdMob** integration to **generate revenue**, strategically deploying **Interstitial & Banner Ad** placements and conducting **A/B testing** on ad frequency, achieving a 15% increase in CTR (click-through rate).

Generating Question Paper from Handwritten Notes using T5 Transformer

- Automated the process of question paper generation from textual / handwritten notes, by combining **OCR** using **Google Cloud Vision** from GCP with a **T5 Transformer** model fine-tuned for **text summarization** and **question generation**.
- Built a **Python** system to enable generation of MCQ/short-answer/True-False questions, allowing educators to upload scanned notes and download formatted question papers (PDF/Word) in < 2 minutes.

Anomaly Detection in Multi-variate Time Series Data using Facebook Prophet

([Research Paper](#))

- Developed a forecasting model for **Anomaly Detection in Time Series Data** using **Unsupervised Machine Learning** techniques and '**Facebook (Meta) Prophet**' algorithm comparing our predictions to **identify outliers**.
- Optimized model accuracy to 96.7%** through **iterative tuning** to correctly identify anomalies in online advertisement KPIs focusing on cost-per-click (CPC) and cost-per-thousand impressions (CPM).

Text Summarization of Traditional Indian Languages using Self-Attention Mechanism

([Research Paper](#))

- Headed & authored research on **Transformer** based **abstractive text summarization** for low-resource regional Indic languages.
- Architected a **self-attention mechanism** optimizing **positional encoding** for **morphologically rich languages**, to outperform existing **sequential model IndicBART** in **ROUGE scores** and achieving **BLEU scores** of 0.62 on a 1,000+ newspaper dataset.